Does monthly threshold updating catch more fraud within a fixed review budget?

Eray Gulerler | Independent research | Warsaw, Poland

Abstract

Across six held-out months of simulated card transactions, the monthly cutoff caught 330 of 2,012 fraudulent transactions (16.40%) and the fixed cutoff caught 302 (15.01%). Both used the same frozen model and could spend one review per 1,000 transactions processed. Monthly updating used 362 reviews and the fixed cutoff used 331. The paired recall difference was +1.39 percentage points (95% block-bootstrap interval 0.74 to 2.08).

Matching realized review counts retrospectively gave 301 frauds caught with monthly updating and 301 with the fixed cutoff across 330 reviews per policy. Monthly updating used more of the available capacity. We observed no gain at equal realized workload in this simulation.

Question and baseline

A fraud investigator can review only a limited number of alerts. A score cutoff set in June may produce too many referrals in a busy month or leave review capacity idle in a quiet one. We ask whether resetting that cutoff using last month's scores improves transaction-level fraud recall under the same limit on reviews.

The preceding study [1] selected a random forest trained with SMOTE and held-out sigmoid calibration. It achieved 0.7547 average precision and 0.6923 recall at its validation-selected cutoff on a chronological ULB/Worldline holdout. The underlying dataset spanned only two days. Here the model recipe carries over to a two-year simulation with different features. The earlier recall and this study's recall measure different operating points on different populations.

Prior work

Dal Pozzolo et al. [2] describe how investigator capacity, changing fraud patterns and delayed feedback affect evaluation. Le Borgne et al. [3] evaluate the highest-ranked cards each day under a fixed investigation capacity. Here an incoming transaction must cross a cutoff estimated from past scores, and the decision to review it must be feasible at its time of arrival.

Data and chronological design

The Sparkov release [4, 5] provides two files covering January 2019 through December 2020. We pooled them, checked all transaction IDs for uniqueness, then ordered by timestamp and ID. The table gives transactions and labeled frauds for every month. All six test months cleared the protocol's minimum of 100 frauds and 35 available reviews.

Month	2019 transactions	Frauds	2020 transactions	Frauds
Jan	52,525	506	52,202	343
Feb	49,866	517	47,791	336
Mar	70,939	494	72,850	444
Apr	68,078	376	66,892	302
May	72,532	408	74,343	527
Jun	86,064	354	87,805	467
Jul	86,596	331	85,848	321
Aug	87,359	382	88,759	415
Sep	70,652	418	69,533	340
Oct	68,758	454	69,348	384
Nov	70,421	388	72,635	294
Dec	141,060	592	139,538	258

One model, four consecutive periods

We trained the preceding study's 100-tree random forest (depth 12, minimum leaf size 2, square-root feature sampling, seed 42) on 2019 alone. Predictors were log amount, one-hot merchant category, geographic distance, and sine/cosine encodings of hour and weekday. Training-only scaling and SMOTE (minority-to-majority ratio 0.1, five neighbors) preceded fitting. January through May 2020 supplied sigmoid calibration at the observed prevalence. No training or calibration updates occurred thereafter.

June 2020 set both starting thresholds. At each boundary we took the score at rank floor(reference volume / 1,000), resolving ties with the smaller transaction ID. The fixed policy retained June's boundary. The monthly policy used only scores from the preceding calendar month. July through December supplied paired test observations. July therefore serves as a check that identical thresholds give identical decisions.

When a review slot becomes available

After transaction i the allowance is floor(i / 1,000), resetting at the month boundary. A qualifying transaction uses an available slot immediately. A crossing with no available slot expires. This rule uses neither the final month volume nor future scores. Primary recall divides reviewed frauds by all frauds in the six test months.

Fraud caught and workload

Table 1. Fixed / monthly pairs. Alerts are all crossings before capacity enforcement. An asterisk identifies demand above that month's allowance. Reviews and caught frauds are realized counts.

2020	Frauds	Slots	Alerts (F / M)	Reviews (F / M)	Caught (F / M)
07	321	85	60 / 60	60 / 60	54 / 54
08	415	88	54 / 75	54 / 73	48 / 66
09	340	69	71* / 114*	60 / 69	57 / 65
10	384	69	63 / 61	52 / 51	50 / 49
11	294	72	68 / 69	62 / 62	56 / 56
12	258	139	43 / 47	43 / 47	37 / 40
Total	2,012	522	359 / 426	331 / 362	302 / 330

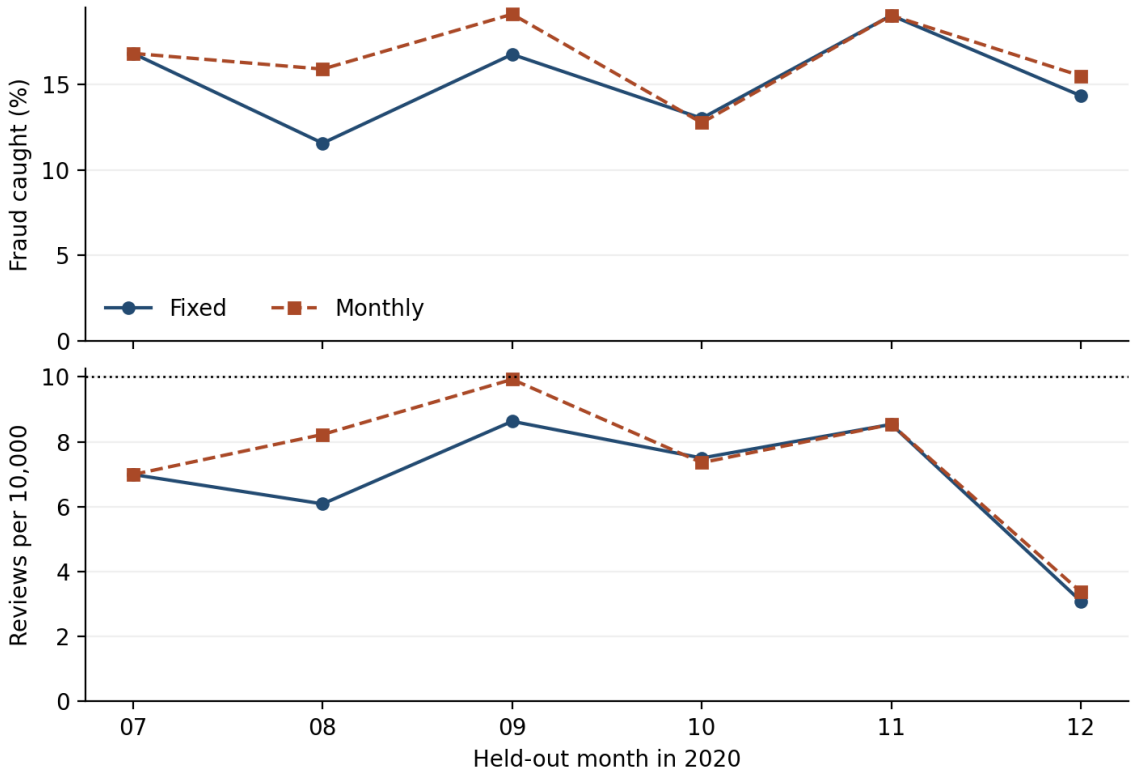


Figure 1. Transaction-level fraud recall (upper panel) and actual reviews per 10,000 incoming transactions (lower panel). July's series coincide because both policies use June's cutoff.

In September, both policies generated more alerts than the 69 available slots. The fixed cutoff produced 71 alerts and the monthly cutoff produced 114. Neither policy exceeded the cap on completed reviews in any month. Across 525,661 test transactions, the fixed policy used 6.30 reviews per 10,000 (95% interval 5.51 to 7.07) and monthly updating used 6.89 (6.02 to 7.78).

Interpretation and limits

Uncertainty and equal review volume

The fixed cutoff caught 15.01% of fraud (95% interval 13.52 to 16.58). Monthly updating caught 16.40% (14.78 to 18.10). The paired monthly-minus-fixed interval was 0.74 to 2.08 percentage points. We resampled circular blocks of seven calendar days within each month, using the same draws for both policies, 5,000 replicates and seed 42. The percentile intervals condition on the fitted policies and observed decisions. They describe day-to-day variability within this calendar.

The common allowance left room for different spending. Matching completed reviews within each month gave 330 reviews per policy and recall of 14.96% for both. August accounted for 18 of the 28 additional frauds caught under the common allowance. Its advantage disappeared when each policy was limited to its first 54 completed reviews. Matching is retrospective because the smaller count becomes known only when the month closes.

What the comparison can establish

The generator encodes spending and fraud profiles. The ranking can exploit those profiles more readily than it might exploit fraud patterns at a bank. Synthetic labels are complete, repeat-card transactions are correlated, and a review has no effect on later transactions. Each fraudulent transaction counts once regardless of loss or whether the same card already triggered a review.

Only five months allow the policies to differ. The block intervals omit uncertainty from model fitting, other simulation draws and longer-term changes. A different review budget, allowance schedule, label process or scoring model could change the result. The fixed ranking also means that a monthly cutoff changes where and when reviews are spent, without improving score ordering.

Reproduction

From the repository root, after installing the pinned study dependencies, run

python -m research.monthly_thresholds.run

Code, data audit, thresholds, intervals and exact environment | Technical report

References

[1] Gulerler, E. (2026). Rare-event classifiers under distribution shift.

[2] Dal Pozzolo et al. Credit Card Fraud Detection: A Realistic Modeling and a Novel Learning Strategy. IEEE Transactions on Neural Networks and Learning Systems.

[3] Le Borgne et al. Reproducible Machine Learning for Credit Card Fraud Detection. Baseline fraud detection system.

[4] Kartik2112. Credit Card Transactions Fraud Detection Dataset. Kaggle.

[5] Sparkov Data Generation. Simulator source and transaction profiles.